

Rahul Ray

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MSc Data Science student at Leiden with a habit of building things to answer questions I couldn't find clean answers to. Interested in what actually happens when you push a method past its assumptions.

Education

MSc Computer Science – Data Science *Leiden University, Netherlands* Sep 2025 – Aug 2027
BTech Computer Science *SRM Institute of Science and Technology, Chennai* 2021 – 2025

Work Experience

Machine Learning Intern *Mar 2025 – Aug 2025 · Fox Solutions Pvt. Ltd., India*
• Developed a computer vision pipeline using Vision Transformers (ViT) end-to-end, from model training through to edge deployment with ONNX, optimising for inference latency and memory footprint on constrained hardware.

Projects

Can LLMs Make Tabular ML Better? *writeup*  [GitHub](#)
Python · XGBoost · LLaMA 3.1 · sentence-transformers · Optuna · SHAP

- Tested whether LLM-generated semantic features improve XGBoost on tabular data using a rigorous nested CV pipeline (3×5-fold, leak-free PCA, Optuna tuning) with Wilcoxon tests and Cohen's d for evaluation.
- The hypothesis was not supported on either dataset — documented the negative result honestly, including SHAP analysis showing LLM features added noise rather than signal.

AutoInsight – End-to-End AutoML Web App *writeup*  [GitHub](#)
Python · Streamlit · XGBoost · LightGBM · CatBoost · SHAP · Optuna · Groq API

- Browser-based AutoML tool that trains 7 models in parallel with stratified k-fold CV, SHAP explainability, and Bayesian tuning; live on Hugging Face Spaces.
- Privacy-preserving by design: the LLM receives only aggregated metrics, never raw user data; responses stream live via Groq and are exportable as JSON.

Quake – Learned Heuristic A* on BSP Geometry *writeup*  [GitHub](#)
Python · PyTorch · XGBoost · NetworkX · BSP v29 parsing

- Parsed Quake 1's raw BSP format from scratch to build navigation graphs across all 38 maps; generated 625k ground-truth pathfinding pairs via Dijkstra.
- Trained MLP and XGBoost heuristics that cut A* node expansions by 42% while keeping paths within 2% of optimal, with train/test split by episode to prevent leakage.

Epoch-wise Double Descent Under Label Noise – ResNet-18 / CIFAR-10 *writeup*  [GitHub](#)
Python · PyTorch · ResNet-18 · CIFAR-10

- Reproduced Nakkiran et al. (2019)'s epoch-wise double descent under symmetric label noise (η up to 30%), tracking clean vs. corrupted accuracy separately to identify the mechanistic inflection point.
- Label smoothing degraded performance; Mixup improved final test loss without eliminating the peak — both reported as honest partial results alongside entropy analysis of the memorisation phase.

Newsphere Live – Semantic News Map *writeup*  [GitHub](#)
Python · sentence-transformers · UMAP · HDBSCAN · Three.js · GitHub Actions

- Self-updating 3D news map where proximity = semantic similarity: fetches 11 RSS feeds every 6 hours, embeds articles via MiniLM, reduces to 3D with UMAP, and clusters with HDBSCAN.
- Fully serverless — rendered with Three.js and published to GitHub Pages via GitHub Actions with zero backend and zero cost.

Technical Skills

ML / DL	PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, ViT, ResNet, SHAP, Optuna
NLP / LLMs	sentence-transformers, Groq API, LLaMA 3.x, UMAP, HDBSCAN
Tools	Python, SQL, pandas, NumPy, Streamlit, Plotly, NetworkX, Git, ONNX, GitHub Actions

Languages

English – C1 Professional Proficiency (*IELTS Certified*) | **Hindi** – Native